

Networking - Day - 15 (Assignment: 15)

Task 1: DNS - How Names Become IP

1 What happens when you type google.com in a browser?

→ When I type a website like google.com in a browser my system asks a DNS Server to convert the name into an IP address. Once the IP is found the browser connects to that server and loads the website.

2 What are these records type? Write any line each?

→ Records types are:

(i) A → Converts a domain to IPV4 address

(ii) AAAA → Converts a domain to IPV6 address

(iii) CNAME → Alias pointing to another domain

(iv) MX → Mail Server record.

(v) NS → Shows the DNS Server for the Domain

3 Run dig google.com - Identify the A record and TTL from the output.

→ The DNS command resolved google.com into IP address. The A record shows website IP and TTL shows how long it will be valid.

Task 2: IP Addressing

1. What is an IPv4 address? How is it structured?

→ An IPv4 address is a numeric address used to identify devices on a network. It is written in dotted format like
192.168.1.10

2. Difference between public and private IP's - Give one example of each.

→ Public IP: Public IP is usable on Internet
example (8.8.8.8)

→ Private IP: Private IP is used inside a local network.
example (192.168.1.10)

3. What are the private IP ranges?

The private IP ranges are

- 10.x.x.x
- 172.16.x.x - 172.31.x.x
- 192.168.x.x

4. Run: `ip addr show` - Identify which of your IPs are private.

→ The system has a private IP on its main network interface, showing it is connected. Other interfaces are for internal and docker communication.

Task 3 - CIDR and Subnetting

- 1
→ What does /24 mean in 192.168.1.0/24
→ The 24 tells about network size - How many device can join
→ /24 means 24 bits are used for network portion, leaving 8 bits for host.
- 2
→ How many usable host in a /24? A/16?
→ A/28?
→ /24 → 254 host
→ A/16 → 65,534 host
→ A/28 → 14 host.
- 3
→ Explain in your own words: Why do we need Subnet?
→ We need Subnet to divide large network into smaller parts so it is easier to manage, reduces traffic, and improves performance.
- 4
→ Quick exercise - fill in?

CIDR	Subnet-Mask	Total IPs	Usable host
/24	255.255.255.0	256	254
/16	255.255.0.0	65,536	65,534
/28	255.255.255.240	16	14

Task-4 : Ports - The Doors to Services.

1 What is a port? Why do we need them?
 → A port is a logical communication that allows multiple services to run on the same IP address.

2 Document these common ports:

Port	Service
22	SSH
80	HTTP
443	HTTPS
53	DNS
3306	MySQL
6379	Redis
27017	MongoDB

3 Run `ss -tulpn` - match at least 2 listening ports to their services?

→ The output shows active service listening on ports like SSH (22) and HTTP (80), confirming that network services are running and accepting connections.

Task 5: Putting It Together

1 You run `curl http://myth.com:8080` -
What networking concept from today
are involved?

→ DNS finds the IP, TCP/IP connects
to the server, and the host 8080
sends the request to the app.

2 Your app can't reach a database at
10.0.1.50:3306 - What would you
check first?

→ I would check network connection,
firewall rules, and if database service
is running on the host.